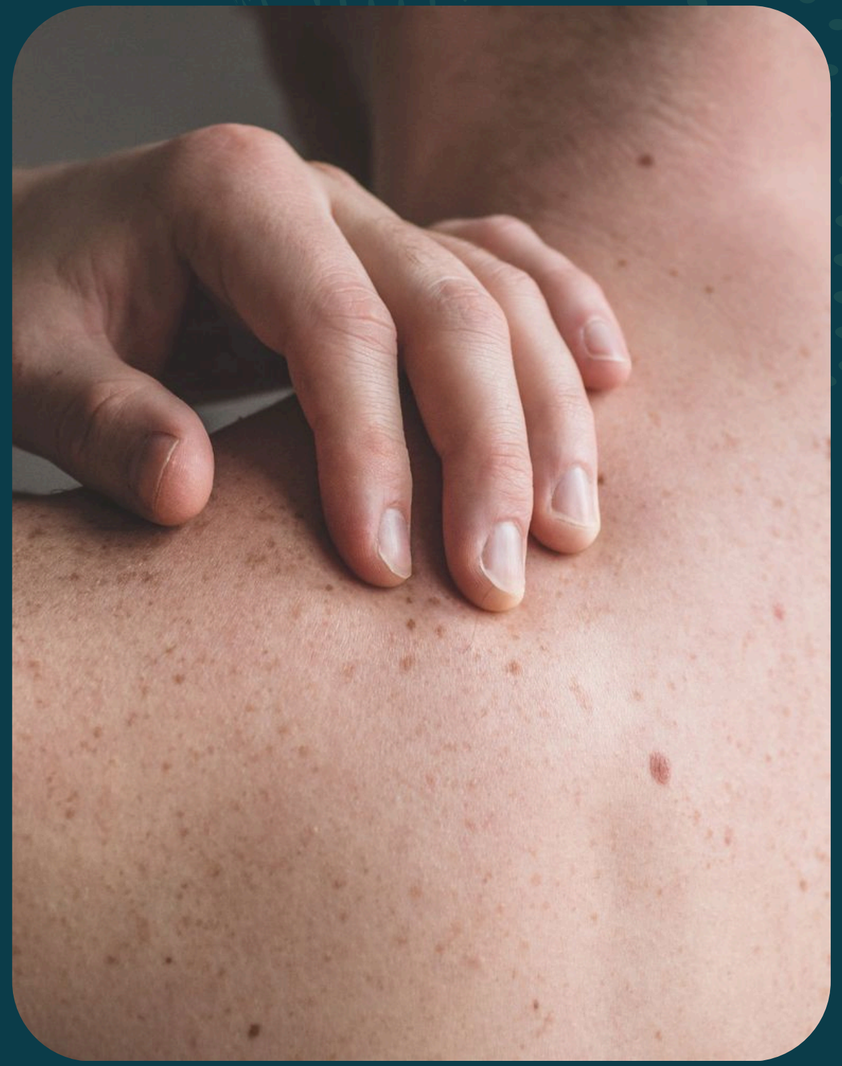
Kirirom Institute of Technology

Skin Lesion Classification

Three deep-learning approaches on a 14-class image dataset



Sokha Marady | Batch 13, Software Engineering
Lecturer: Mr. Soklong HIM | Deep Learning Final Project

Problem and motivation

Kirirom Institute of Technology

Given one photo of skin, predict which of 14 classes it shows.

8

Skin lesion types

Melanoma, melanocytic nevi, basal cell and squamous cell carcinoma, actinic keratoses, benign keratosis-like, dermatofibroma, vascular lesions

5

Viral skin diseases

Chickenpox, cowpox, HFMD, measles, monkeypox

1

Healthy skin

Normal skin, so the model can say "nothing here"

Research question: how much does pretrained knowledge help, compared with training a network from scratch?

Dataset and split

Kaggle dataset ahmedxc4/skin-ds: 36,656 images, 14 classes, official train / val / test folders

29,322
Train (80%)

3,660
Validation (10%)

3,674
Test (10%)

- Loaded with torchvision ImageFolder from the official folders as provided; we did not re-split
- All images resized to 224 x 224; test images use the same preprocessing as validation

Heavy class imbalance

Kirirom Institute of Technology

10,300

Melanocytic nevi
(train images)

about 54x

more images

191

Dermatofibroma
(train images)

Weighted cross-entropy

Rare classes count more in the loss, so the model cannot ignore them.

Macro-F1 as the main metric

Every class counts equally, unlike accuracy, which the big classes dominate.

Three approaches compared

Kirirom Institute of Technology

1

Custom CNN from scratch

392,974

trainable parameters

Small CNN, random start. Learns everything from skin images only.

2

ResNet-18, frozen

7,182

trainable parameters

ImageNet weights kept fixed. Only the final linear layer is trained (linear probe).

3

ResNet-18, fine-tuned

11,183,694

trainable parameters

ImageNet weights as the starting point. Every layer is trained at a low learning rate.

Experimental setup

Kirirom Institute of Technology

Setting	CNN scratch	ResNet frozen	ResNet fine-tuned
Learning rate	1e-3	1e-3	1e-4
Weight decay	1e-4	1e-4	1e-4
Max epochs	25	15	15
Training time (min)	160.6	98.0	99.9

Shared by all three

- Adam optimiser
- ReduceLROnPlateau scheduler
- Batch size 32, input 224 x 224
- Weighted cross-entropy, seed 42

Best checkpoint

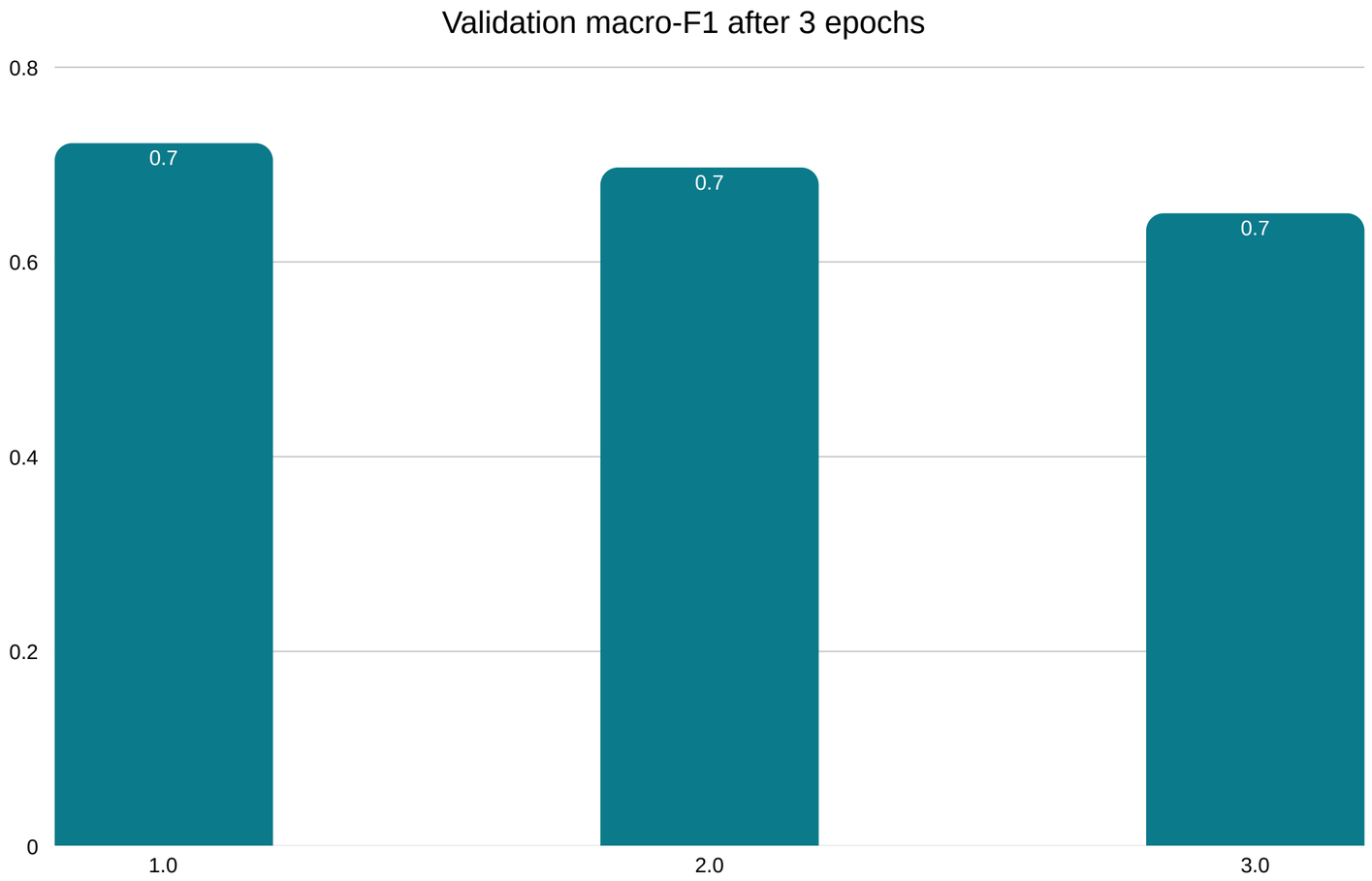
Saved at the lowest validation loss, then evaluated once on the test set.

Metrics and hardware

Accuracy, macro-F1, macro-precision, macro-recall. Tesla T4 GPU, PyTorch.

Hyperparameter tuning

Kirirom Institute of Technology



What we learned

- Best: lr 1e-4 with weight decay 1e-4 (0.722)
- Removing weight decay lowered the score
- A smaller lr of 1e-5 learned too slowly in 3 epochs

Results on the test set

Kirirom Institute of Technology

Approach	Accuracy	Macro-F1	Macro-Prec.	Macro-Recall	Trainable params	Time (min)
1. CNN from scratch	0.5822	0.5205	0.5053	0.6123	392,974	160.6
2. ResNet-18 frozen	0.6184	0.5415	0.5179	0.6074	7,182	98.0
3. ResNet-18 fine-tuned	**0.8419**	**0.8330**	**0.8117**	**0.8607**	**11,183,694**	**99.9**

+22 pts

accuracy vs frozen ResNet

+29 pts

macro-F1 vs frozen ResNet

~100 min

same cost as frozen on a T4

Fine-tuning clearly wins

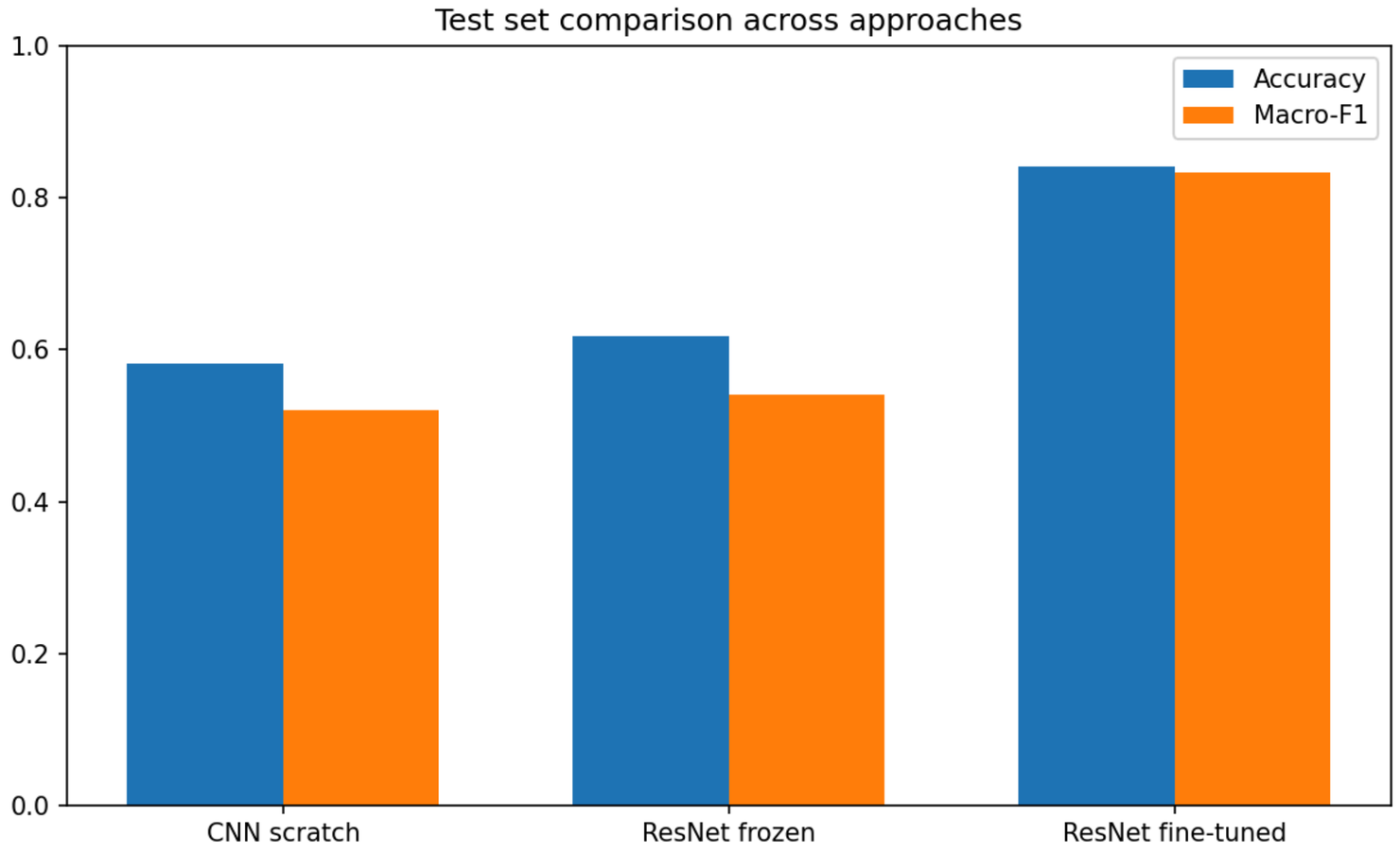
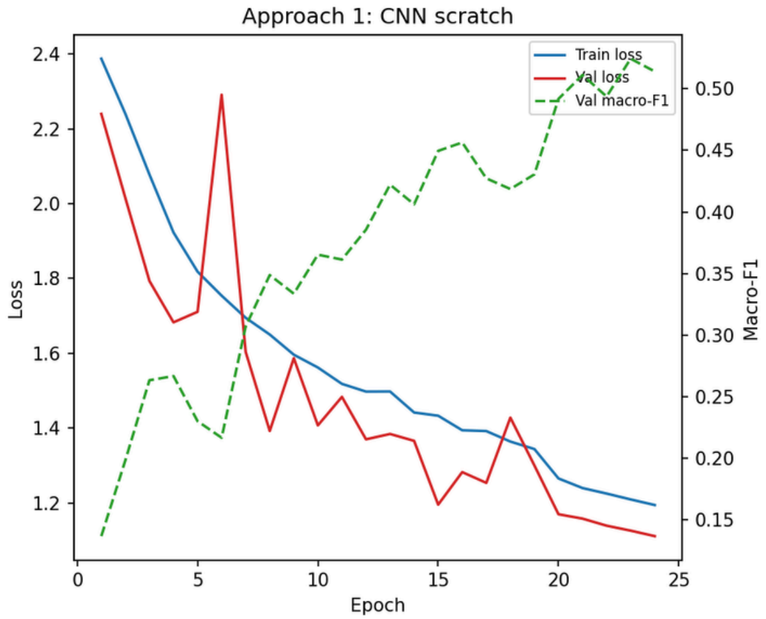


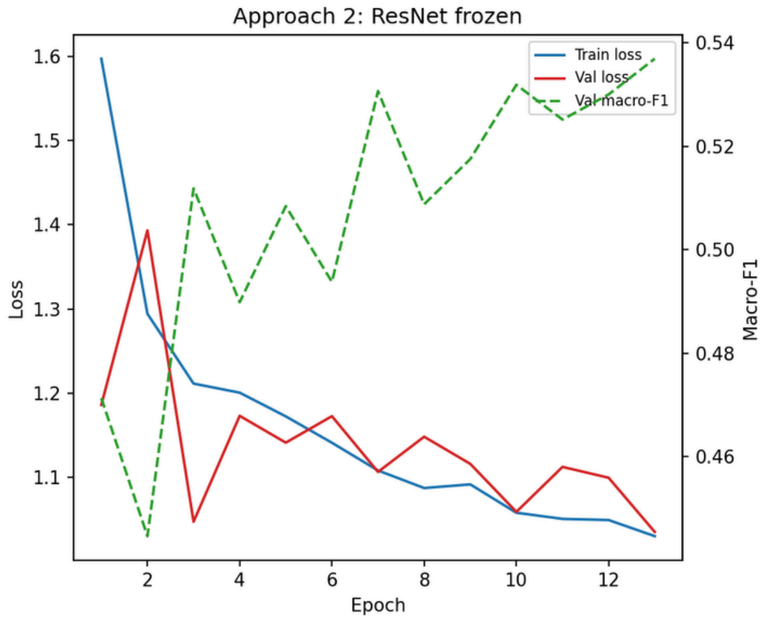
Figure: test accuracy and macro-F1

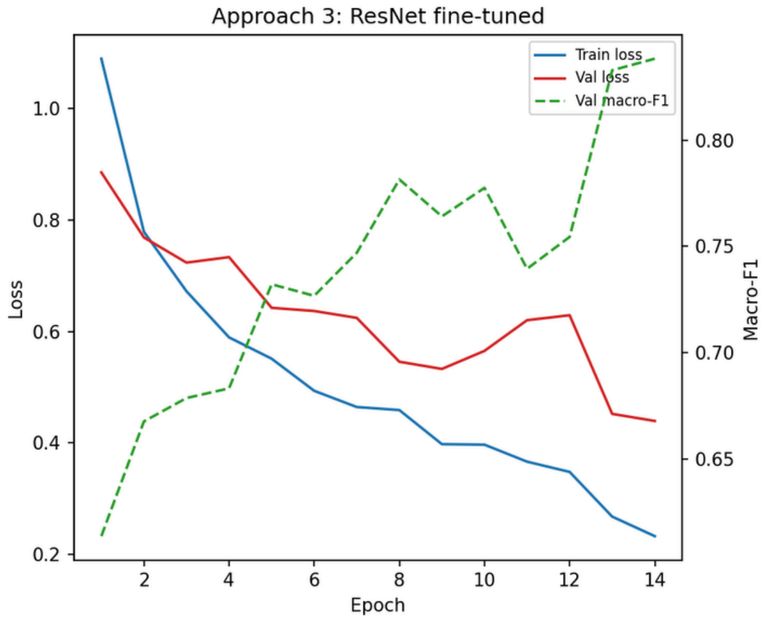
- Fine-tuned: 0.842 accuracy, 0.833 macro-F1
- Frozen and scratch models both stay near 0.6 accuracy
- The scratch CNN was the slowest to train (160.6 min)
- The fine-tuned model has a small gap between accuracy and macro-F1, so rare classes are handled better

Learning curves: overfitting and underfitting

Kirirom Institute of Technology







CNN scratch: underfitting

Loss is still falling at the end, so it is undertrained or capacity-limited.

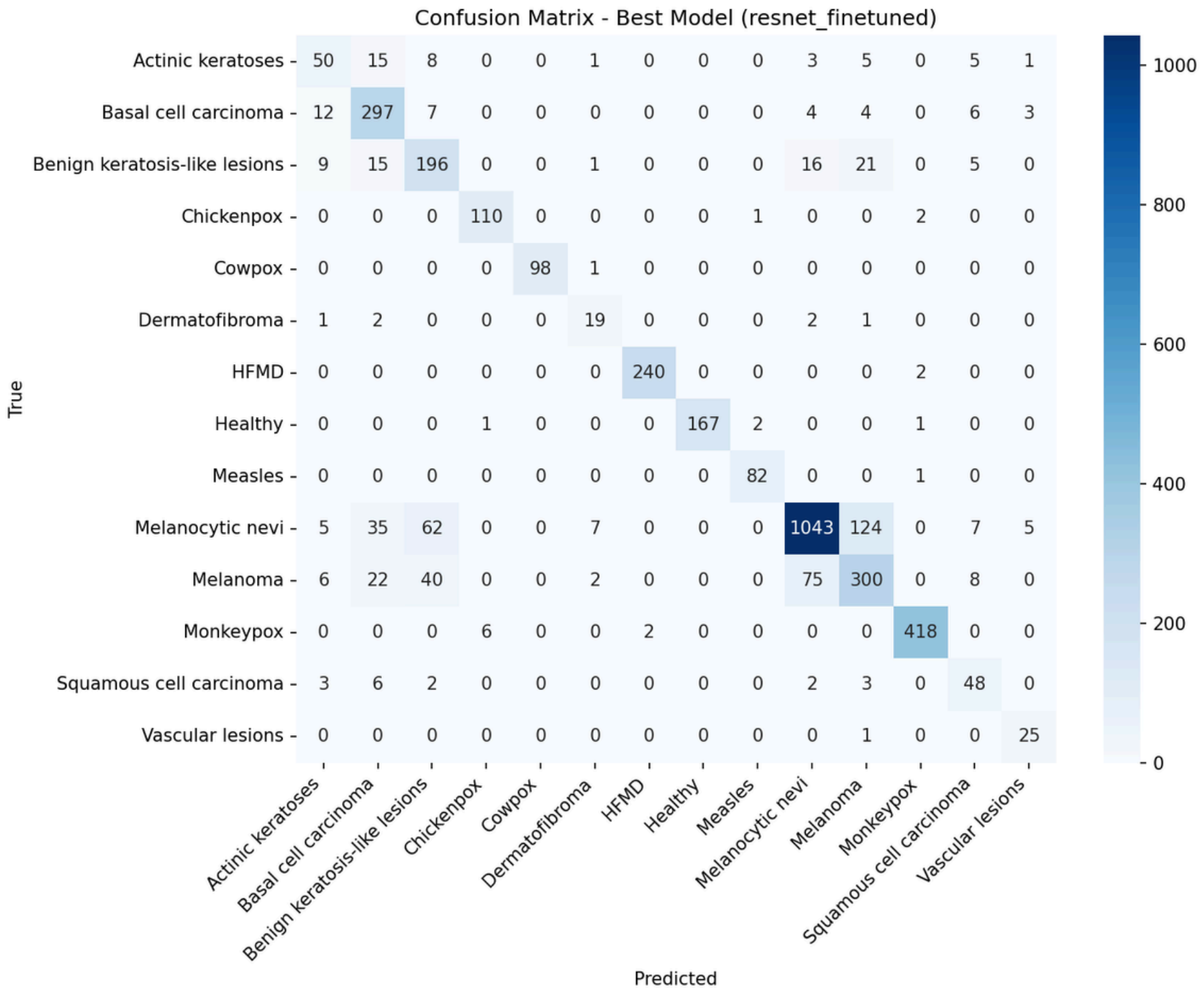
Frozen ResNet: plateau

Plateaus early with a noisy validation curve. Limited by the fixed features.

Fine-tuned: mild overfitting

Train loss 0.23 vs val loss 0.44. Val loss is still lowest at the saved best epoch.

Error analysis: confusion matrix



Nevi and melanoma mix up

124 nevi predicted as melanoma, 75 melanoma predicted as nevi

Actinic keratoses (F1 0.57)

Only 50 of 88 correct; the rest spread to basal cell, benign keratosis, SCC

Viral classes are strong

Chickenpox, cowpox, HFMD, measles, monkeypox and healthy: F1 0.96 to 0.99

Why fine-tuning won, and an informal ensembling check

Kirirom Institute of Technology

Why it won

1

Frozen ImageNet features transfer poorly

Domain gap: skin photos differ from ImageNet, so a linear head on fixed features tops out at 0.618.

2

Every layer adapts to skin images

Pretrained start plus 11.2M trainable weights, versus 393k for the scratch CNN.

3

A tuned, gentle learning rate

lr 1e-4 won the tuning runs and keeps pretrained knowledge intact.

Informal ensembling check

99.62%

fine-tuned model alone, one example photo

56.68%

equal-weight ensemble of all three, same photo

Here the weak CNN diluted the strong model. Not tested on the full test set.

Limitations and future work

Kirirom Institute of Technology

Limitations

- Domain shift: can fail on random internet photos (informal tests), since it learned one dataset's photo style
- Multi-source acquisition bias: image source can leak into what the model learns
- Rare, look-alike classes stay weak: actinic keratoses, melanoma, dermatofibroma, SCC
- Not a diagnostic tool

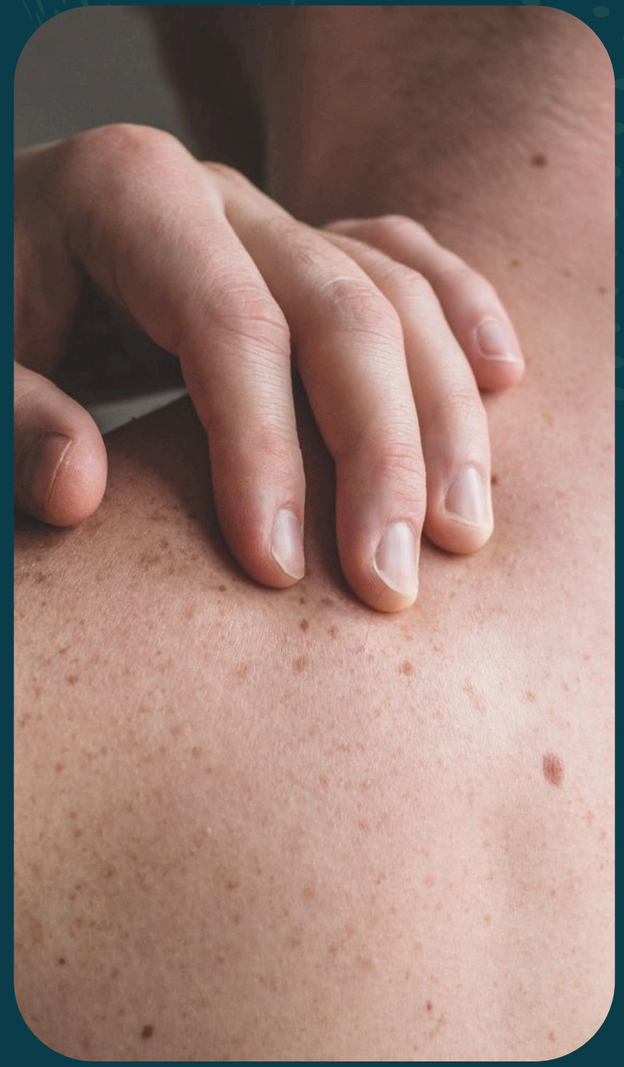
Future work

- Data augmentation and more samples for rare classes
- Weighted or learned ensembling instead of equal weights
- Test on external datasets to measure domain shift
- Try larger pretrained backbones

Kirirom Institute of Technology

Conclusion

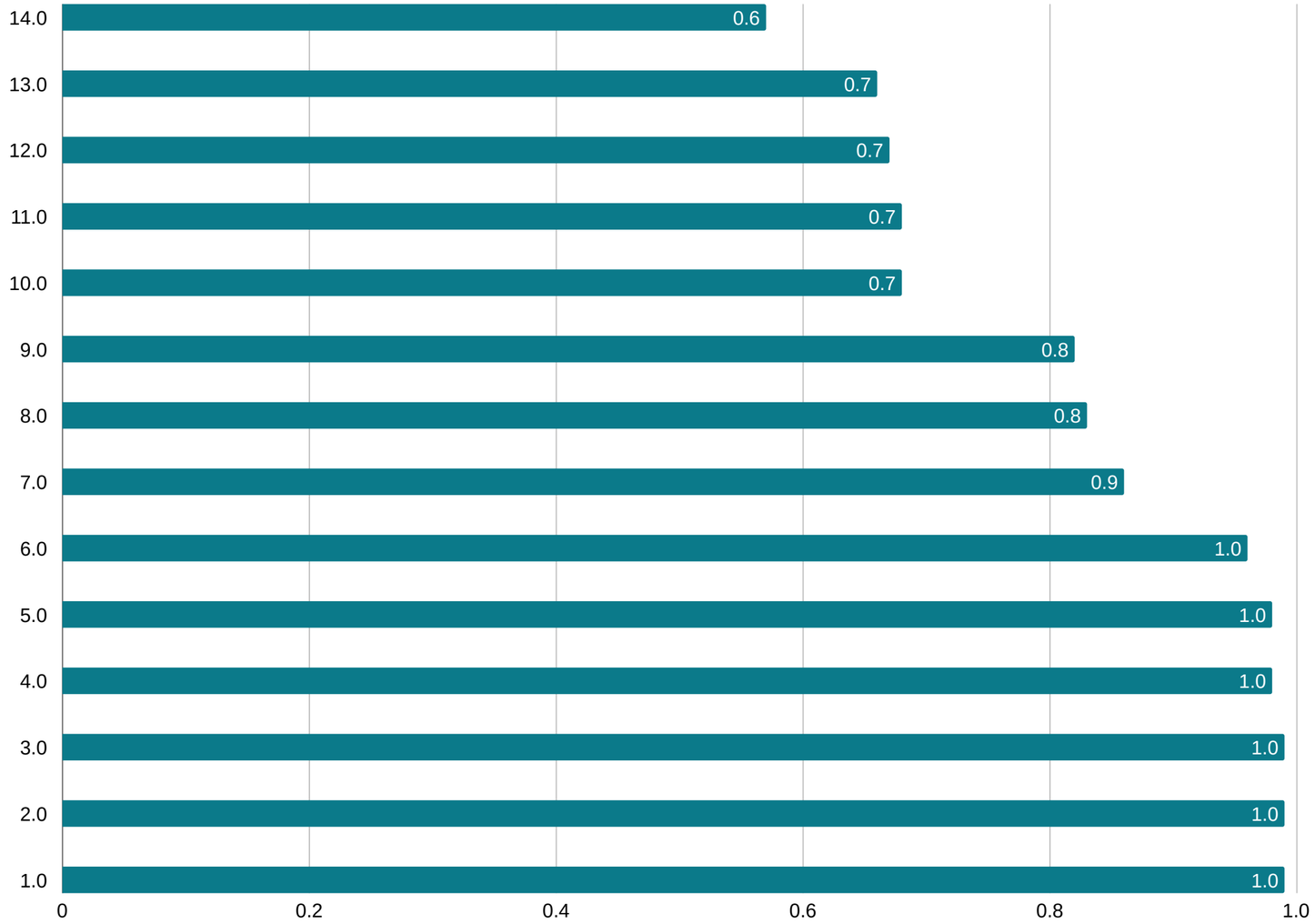
- Fully fine-tuned ResNet-18 reached 0.842 accuracy and 0.833 macro-F1
- Frozen features and a scratch CNN both stayed near 0.6 accuracy
- Main errors: nevi vs melanoma and other look-alike lesions
- Next: rare-class data and external validation



github.com/Smarady67/skin-lesion-classification

Appendix A: per-class F1, fine-tuned ResNet

Kirirom Institute of Technology

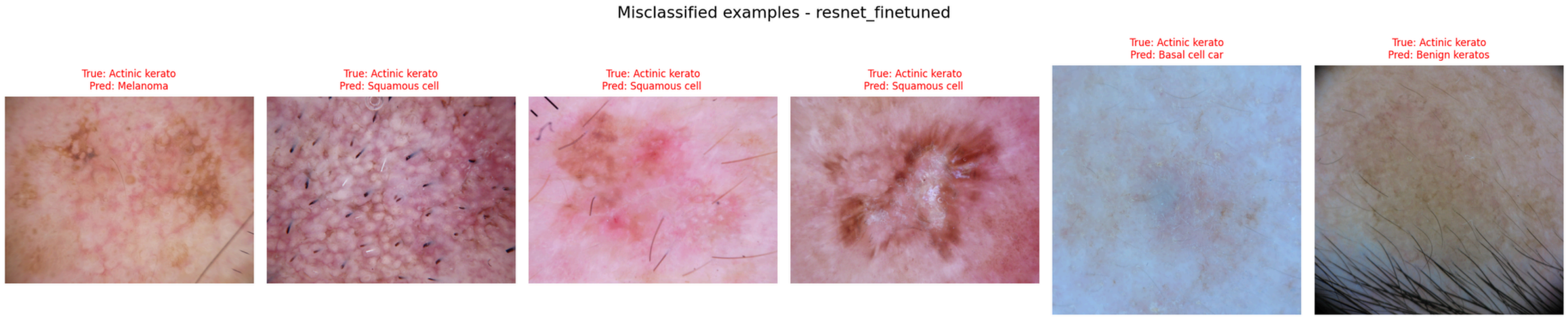


Reading the chart

- Coral: F1 below 0.70
- Weak classes are rare and look alike
- Viral classes: 0.96 to 0.99

Appendix B: misclassified examples

Kirirom Institute of Technology



- All six shown are actinic keratoses, predicted as melanoma, squamous cell carcinoma, basal cell carcinoma or benign keratosis
- These classes share colour and texture, so even a strong model confuses them